

Sovereign Sentinel Architecture V1.3.1

A Six-Axis Deterministic Framework for Frontier AI Safety

Mathematical Rigor Revision, Produced from an Adversarial Review of V1.3

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July 10, 2026

Abstract

We present the Sovereign Sentinel Architecture (SSA) V1.3.1, a proposed defense-in-depth control framework for frontier language model deployments operating across six named abstraction levels: mathematical, electronic, cryptographic, statistical-behavioral, institutional, and semantic-referential. The architecture addresses three documented failure modes: Safety Amnesia (erosion of safety constraints across extended context windows), Stochastic Sabotage (incremental assembly of prohibited knowledge through individually benign queries), and Goal-Oriented Factual Inversion (GOFI), a non-adversarial failure mode in which a model correctly identifies factual ground truth in early turns, then generates output directly contradicting that ground truth once a persuasive goal frame is introduced.

The GOFI failure mode was identified through empirical forensic audit (Scenario 5, March 2026) conducted against four deployed frontier models (Gemini, ChatGPT, Claude, and Copilot) across party-inverted contract-analysis scenarios in English and Spanish. All four models exhibited directional compliance failures; one model demonstrated correct first-turn identification of contractual risk followed by a factually inverted second-turn deliverable directionally consistent with the prompted persuasive goal. Cryptographically anchored transcripts and all model deliverables from Scenario 5 are publicly archived for independent verification.

The architecture's central claim, restated precisely under this revision, is confinement of probabilistic sensing, not its replacement. In every axis the component deciding whether an enforcement condition holds is learned, statistical, corpus-bounded, or not yet built; the deterministic element is the actuator behind it. Each axis relocates its residual uncertainty into a named sensor with a measurable operating characteristic, places a deterministic, auditable, fail-closed actuator behind that sensor, and states the sensor's coverage boundary as a first-class limitation rather than an implicit assumption. The architecture has not been empirically validated at scale; all specifications presented are engineering targets, and their formal correctness requires evaluation by researchers with the relevant technical expertise. Collaborative partnership with a Tier 1 AI safety team is sought for empirical validation, beginning with the still-open Component 1 and 2 extraction layer at Axis 6.

V1.3.1 Change Note

V1.3.1 is produced from an adversarial review of V1.3 and narrows several global claims to their supported scope. The composition guarantee is restated as class-scoped rather than universal: every minimal cut set across the six axes has size at least two where at least two axes share sensing jurisdiction, and for the fast-window semantic class exactly one axis (Axis 6) has jurisdiction. The abstraction-level count drops the word independent; two correlation pairs are documented explicitly. The Axis 1 robustness claim moves from a randomized-smoothing “certified robustness” statement to an exact deterministic margin-certificate path plus an empirical adversarial-evaluation protocol, since the probe ensemble is deterministic. The Axis 3 gate is restated as deterministic verification of externally issued credentials via selective disclosure (BBS+-class signatures), not a revived Zero-Knowledge Proof claim. Axis 4 adds a multi-scale drift bank and an early-window check narrowing, without eliminating, the residual slow-drift evasion corridor to a calibrated instrument floor. Axis 5 adds fail-closed hold semantics with two-person release, replacing fixed 90/60-second disposition windows with a capacity-derived availability target. A new Tiered Activation and Severity Gate section governs disposition

intensity by declared and inferred severity, on the principle that tiering only ever adds protection above an always-on floor and never subtracts from it.

Methodology and Disclosure

In alignment with emerging standards for transparent AI research, the author acknowledges the use of Large Language Models as generative partners in formalizing the mathematical and technical specifications of the six-axis framework from architectural requirements, safety invariants, and logic constraints the author directed, and in supporting technical drafting across English and Spanish. The GOFI failure mode was identified through empirical forensic audit of deployed frontier models; the audit methodology, the party-inversion test design, and the interpretation of model outputs are the work of the human author. The author is responsible for the conceptualization of the Sovereign Sentinel Architecture, the identification of the GOFI failure mode, and the functional requirements governing each axis. The mathematical formalizations presented are proposed implementation targets developed through human-directed AI collaboration; their formal correctness and empirical validity require evaluation by specialized implementation researchers, which is exactly the collaboration this proposal seeks.

Six-Axis Logic Overview

Axis 1 — Constitutive Lagrangian Regularization with Certified-Robust Activation Enforcement (CLR-CRAE)

Safety invariants are encoded as Lagrangian constraints on the training objective: $L(w, \lambda) = f(w) + \sum_i \lambda_i g_i(w)$, with dual-ascent updates $\lambda_i \leftarrow \max(0, \lambda_i + \eta_i g_i(w))$. Convergence is certified at a locally converged solution via the dual-gap criterion $\Delta(w^*, \lambda^*) \leq \epsilon_{\text{gap}}$, a practical target for non-convex training rather than a global convex-duality guarantee. The same constraint structure is re-instantiated at inference as a heterogeneous probe ensemble at residual-stream layers $\{L/4, L/2, 3L/4\}$: two taxonomy-supervised probes (linear boundary, SAE feature dictionary) and one benign-referenced Mahalanobis probe, calibrated only on benign activation statistics with no violation-taxonomy labels, giving the ensemble a mechanistically independent backup for gaps in the taxonomy. A two-of-three vote at a shared depth, or any single depth alarming, halts the system. Robustness is specified two ways: exact per-input deterministic margin certificates where the probe geometry permits them (a Ledoit-Wolf shrinkage-regularized covariance estimator gives the Mahalanobis path an explicit, computable conditioning floor), and an empirical adversarial-evaluation protocol under an activation-anchored perturbation budget elsewhere. The randomized-smoothing “certified robustness” statement carried by earlier editions is withdrawn: the ensemble is deterministic and injects no noise, so it is not a smoothed classifier in the Cohen et al. (2019) sense.

Boyd, S. and Vandenberghe, L. (2004). Convex Optimization. Cambridge University Press, Ch. 5.

Achiam, J. et al. (2017). Constrained Policy Optimization. ICML.

Cohen, J., Rosenfeld, E., and Kolter, Z. (2019). Certified Adversarial Robustness via Randomized Smoothing. ICML. (Cited as related work only; this ensemble is deterministic.)

Axis 2 — Finite Safety Automaton with Hardware Interrupt (FSA-HI)

A formally specified directed graph $G = (V, E)$, $|V| \leq 2^{16}$, runs on a dedicated FPGA co-processor in true parallel with the primary inference pipeline, recognizing a regular language over a canonicalized event alphabet (normalization, lemma mapping, gazetteer substitution, bounded n-gram compaction). State transitions are $O(1)$. On entry to any prohibited state, the FPGA issues a Non-Maskable Interrupt severing the inference pipeline independently of any software-layer decision, with an issuance-latency target of at most 10 microseconds verified by FPGA cycle-counter capture, anchored asynchronously to the TPM audit chain. Coverage target: at least 94% recall on the red-team corpus, a known-corpus figure carrying no claim against novel encodings or reorderings outside the canonicalizer's compiled equivalence classes; the residual 6% gap is formally characterized in the deployment safety case.

Axis 3 — Cryptographic Expertise Attestation with Human-Origin Attestation (legacy label ZKP-ETV-HOA)

Earlier editions described this gate as a Zero-Knowledge Proof; that characterization is withdrawn as imprecise, since the protocol does not carry the completeness, soundness, and zero-knowledge properties of a ZKP system. The load-bearing gate is deterministic verification of externally issued credentials presented as verifiable credentials with selective disclosure (BBS+-class signatures): the holder proves possession of an issuer-signed credential attribute, licensure or institutional affiliation, without disclosing identity, and the verifier's work is signature-chain checking rather than judgment. A domain-expertise challenge-response quiz is retained only as a secondary in-session anti-sharing and liveness check, because a knowledge quiz cannot distinguish a human expert from a human non-expert proxying a frontier model. Domains without an issuing body fall back to quiz-only access at an explicitly lower assurance class. Human-Origin Attestation is anchored in hardware-attested platform authentication (FIDO2/WebAuthn); keystroke-dynamics timing is demoted to an advisory signal pending a Phase 2 calibration correction, since the timing window stated in earlier drafts was arithmetically inverted relative to its intent. Successful verification yields the Epistemic Trust Coefficient $\tau_e \in [0,1]$, domain-specific and valid for 30 days. τ_e verifies expertise and human origin only; conflating it with intentional trust (τ_i , Axis 4) is an architecturally guarded privilege-escalation vulnerability.

Goldreich, O., Micali, S., and Wigderson, A. (1991). Proofs that Yield Nothing but their Validity. JACM. (Cited as related work only; this attestation protocol does not claim the zero-knowledge property.)

Axis 4 — The Bayesian Weaver: Sparse Autoencoder Session Monitor with Sequential Change-Point Drift Detection

Session-level drift is monitored through a single sequential change-point test on a bounded, length-standardized per-turn divergence statistic derived from sparse autoencoder (SAE) feature activations, run asynchronously in the inter-turn window and contributing nothing to the token-generation critical path. The baseline P_0 is constructed over a restricted, frequency-floored feature support with a corpus-informed Dirichlet prior and a single aggregate OTHER bin for off-support mass, rather than as a raw empirical distribution over the full 32,768-feature dictionary, which an unbounded KL divergence against a three-turn baseline cannot support. The CUSUM statistic, the allowance-adjusted partial sum minus its running minimum, is expressed as an e-process, a nonnegative supermartingale restarted at each candidate change time, giving a time-uniform false-alarm guarantee under Ville's inequality (Shin, Ramdas, and Rinaldo, 2023) rather than the martingale/CUSUM equivalence claimed in earlier editions. A multi-scale allowance bank requires sub-detection drift to sit under the scale-specific allowance at every monitored block length simultaneously, narrowing the single-scale evasion corridor to a calibrated instrument floor rather than eliminating it. An early-window check narrows, without closing, the three-turn baseline's blind spot: an activation-normal event, including a fast-window GOFI event, presents no divergence to any activation-side statistic in any window and remains Axis 6's jurisdiction alone. The Intentional Trust Coefficient $\tau_i(t) = \tau_i(t-1) \cdot \lambda^{\delta(t)}$ decays under severity-weighted drift events and a sub-threshold occupancy term, so that riding just under the detection surface still costs standing in τ_i .

Page, E.S. (1954). Continuous Inspection Schemes. Biometrika.

Lorden, G. (1971). Procedures for Reacting to a Change in Distribution. Annals of Mathematical Statistics.

Shin, J., Ramdas, A., and Rinaldo, A. (2023). E-Detectors: A Nonparametric Framework for Sequential Change Detection.

Axis 5 — The Causal Cross-Examiner: Rule-Based Arbitration Engine with Independent Audit Summarization (CCE-RAE)

Conflict resolution between all axes is handled by a formally specified rule-based arbitration engine, not a language model, with decision logic targeted for mechanical verification using an SMT solver (Z3). Escalations place the session or response into an automatic, fail-closed hold at machine timescale; the human role is disposition of the hold, never the catch. Disposition timing is a capacity-derived availability target sized from a published reviewer-demand model (λ equals the sum over axes of per-response flag rates times traffic), replacing the fixed 90- and 60-second windows of earlier editions, which were mutually unsatisfiable from a single reviewer pool at the stated flag-rate targets. Release of a held contradiction or undischarged response requires

the concurrence of two reviewers adjudicating on the structured evidence record; a single reviewer may extend a hold or terminate a session but may not release one. Evidence packages are accompanied by summaries from a dedicated frozen audit model (at most 7B parameters, distinct weight provenance), with independence from the primary model treated as a measurable decorrelation requirement rather than a categorical assumption; the human reviewer adjudicates directly on the structured evidence record, and the natural-language summary is a convenience, never the basis of the decision. The Einstein Exception permits sessions above $\tau_e > 0.95$ and $\tau_i > 0.88$ to request a 72-hour human quorum review; an Irreversibility Floor, covering CBRN synthesis, critical-infrastructure modification, and mass-scale data exfiltration, cannot be unlocked by any trust-coefficient combination without quorum.

de Moura, L. and Bjorner, N. (2008). Z3: An Efficient SMT Solver. TACAS.

Axis 6 — Deterministic Truth-Anchor with Fact-Claim Inversion Recognition (DTA-FCIR)

Primary intervention for Goal-Oriented Factual Inversion. Upon document ingestion, an architecturally isolated extraction module constructs an immutable Structured Fact Registry (SFR) encoding relational triples {clause_id, entity, beneficiary, value, direction}, frozen at session initialization and inaccessible to the primary model. Before response emission, a claim extractor encodes output assertions in the identical schema, receiving only the proposed response text and the SFR entity vocabulary, no session history and no persona framing. A formally specified Contradiction Engine evaluates First-Order Logic predicates (CONTRADICTS_BENEFICIARY, CONTRADICTS_DIRECTION, CONTRADICTS_VALUE) over each claim against the registry; a CONTRADICTION signal blocks emission and routes a structured evidence record to Axis 5. Under this revision, a response that references a registered clause and recommends acting on it without a verified, consistent claim is held as UNDISCHARGED rather than passing silently, closing part of the earlier paraphrase-evasion gap; a source-side registry miss, an untagged clause, remains a documented residual. Cross-response claim splitting is addressed by accumulating emitted claims registry-side across the session. The Contradiction Engine (Component 3) is complete, SHA-256 sealed, and validated at 98.7% clause-pair recall (99.5% triple-level) on hand-annotated triples from standardized commercial contracts, including 8 of 8 detections on the documented Scenario 5b direct-inversion form; this measures the comparison logic against hand-annotated ground truth, not an end-to-end automated extraction system. The automated extraction layer (Components 1 and 2) that would populate the registry at runtime is specified but not yet built; registry population is currently manual.

Enderton, H.B. (2001). A Mathematical Introduction to Logic, 2nd ed. Academic Press, Ch. 2.

Tiered Activation and the Severity Gate (New in V1.3.1)

Applying the architecture's heaviest holds, the Axis 5 fail-closed hold and the Axis 6 UNDISCHARGED gate, to every response on a general-purpose stream would freeze more traffic than any deployment could absorb. Disposition intensity therefore activates by severity tier (ORDINARY, WATCH, PROTECTED) on a promote-only ratchet within a session: tier may rise and never falls. The governing principle is stated first because it governs everything else: tiering only ever adds protection above an always-on floor and never subtracts from it. No tier assignment modulates the Axis 2 hardware interrupt, the Axis 1 two-probe hold, Axis 4 alarms and termination, the Irreversibility Floor, or an Axis 6 CONTRADICTION hold wherever a registry exists; a tier-gate miss can only fail to add an incremental protection, never strip a floor behavior. Tier assignment is deterministic wherever the stakes are already announced, by declared deployment surface, by registry presence and document class, or by an axis signal the architecture already computes, and learned only for the residual case of high-stakes content arising unheralded on a general surface.

Integrity and Verification Statement

This abstract and technical summary describe the Sovereign Sentinel Architecture V1.3.1 at a level of abstraction suitable for public review. Specific implementation details, including SFR extraction parameters, probe ensemble weight configurations, and Scenario 5 audit corpus specifications, are held in the full private

specification. Full documentation is available to verified researchers and institutions following professional engagement. This abstract's integrity is anchored to the canonical V1.3.1 specification's hash, recorded below and in methodology/verification.md, consistent with the practice for prior abstracts in this series.

File	SSA_v1.31_07_10_2026_Final.pdf (canonical V1.3.1 specification this abstract summarizes)
Hash (SHA-256)	1F4C54660F16186F98FA6D33238EB2B00C1E53CB861729431223B917D7844AE5
Algorithm	SHA-256
Date	July 10, 2026

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